Balancing Attention to Base and Novel Categories for Few-Shot Object Detection in Remote Sensing Imagery

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***Abstract*— Few-shot object detection (FSOD) has garnered widespread attention in recent years, which makes it possible to learn novel classes with only a handful of labeled samples. Due to the obvious long-tail distribution of remote sensing (RS) data and serious challenges in data labeling, FSOD holds greater practical application value in RS. At present, the FSOD algorithms with fine-tuning of some parameters have attracted much attention due to their stronger incremental learning capacity. However, given large intraclass scale variations and small interclass feature differences of RS objects, we still need to place greater emphasis on the localization and classification of objects. In this article, we introduce the RoI feature refinement (RIFR) method for FSOD in RS imagery, which adopts a powerful training pipeline to better balance attention to the performance of base and novel classes. Aiming at large intraclass scale variations of RS objects, we design a scale-aware feature compensation module (SAFCM). By compensating for insufficient scale information, the model's ability to perceive scale variations of the same class objects has been enhanced. Considering the high similarity among RS classes, we come up with a prototype trihard (PT) loss. It achieves the effect of interclass separability by constraining the relationship between samples and prototypes. Thus, the issue of interclass confusion has been resultfully solved. Comprehensive experiments on three datasets, DIOR, NWPU VHR-10.v2, and FAIR1M-Airplane, can showcase the efficacy of our RIFR method, and it can implement the most outstanding performance currently. The code will be available at: https://github.com/ningerhhh/RIFR.**

***Index Terms*— Few-shot learning, object detection, remote sensing (RS) imagery, RoI feature refinement (RIFR).**

Received 20 June 2024; revised 15 October 2024; accepted 24 November 2024. Date of publication 9 December 2024; date of current version 30 December 2024. This work was supported by the National Key Research and Development Program of China under Grant 2021YFB3900504. *(Corresponding author: Peijin Wang.)*

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Digital Object Identifier 10.1109/TGRS.2024.3512865

I. Introduction

OBJECT detection stands as the central task in interpreting remote sensing (RS) images, and it provides strong support for understanding the surface phenomena [1], [2], [3], [4]. Thus, it is necessary to continuously enhance and develop object detection technology to meet the growing application needs. Conventional object detection methods typically necessitate substantial samples when training deep learning models [5], [6], [7], [8]. However, in the field of RS, the long-tail distribution of object categories is obvious [9], and the labeling process of some samples requires professional knowledge and a large amount of time. It is difficult and expensive to obtain large-scale annotated full-category data for practical applications. Few-shot object detection (FSOD) [10], [11], [12], [13] is introduced to address data-scarcity issues, and it has garnered significant attention and demonstrated remarkable performance.

The objective of FSOD is to effectively transfer the knowledge acquired from base classes to the learning of novel classes with only a few training samples. This task is illustrated graphically in Fig. 1. Concretely, base classes represent the categories with sufficient samples, while novel classes represent the categories that contain a few samples. The overall training process is generally split into base training and few-shot training. Ultimately, it is expected that the model can extend its capability to detect object categories with limited data availability. In general, FSOD methods are classified into two groups: metalearning methods [14], [15], [16], [17] and fine-tuning methods [18], [19], [20], [21]. The methods falling into the first group learn generalized knowledge from numerous small tasks and then adapt swiftly to novel tasks. Most of the FSOD methods in the field of RS adopt metalearning training paradigms, such as FSODM [22], P-CNN [23], and TEMO [24]. They construct the association between support and query samples through metanetworks to detect objects. The second group methods involve pretraining the model with base classes and subsequently fine-tuning it with novel classes. G-FSDet [25] is a representative fine-tuning method for FSOD in RS imagery. By freezing the partial structure, it can maintain the ability of the model to extract features, and effectively overcome the over-fitting problem that is prone to occur during training of novel classes.

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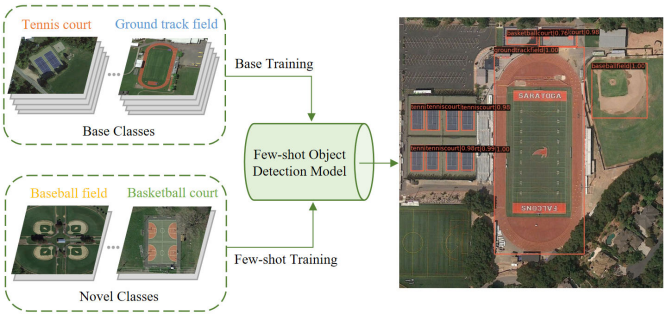


Fig. 1. Description of FSOD task. We need to train an FSOD model using sufficient annotated base classes and a few annotated novel classes. The purpose is to enable the model to achieve decent detection performance on data-limited classes.

Although existing methods achieve excellent performance for FSOD in RS imagery, there remain certain deficiencies. Most methods [14], [15], [16], [18], [19], [20] only focus on the performance of novel classes. Though a few methods [12], [25], [26], [27] can take into account both base and novel classes, they usually design different strategies to predict the base and novel classes, respectively, or require a double-branch design to balance new and old knowledges. In contrast, we hope to design a unified network architecture to enhance the discriminative feature representation across all classes from the source. It can mitigate optimization competition between base and novel classes, ultimately achieving a better balance performance in all classes. In this case, how to optimize the network more appropriately becomes crucial. At the same time, considering that we use RS imagery in our experiments, we will encounter the following more severe challenges.

A. Large Intraclass Scale Variations

Concretely, RS objects belonging to the same class have large differences in scale, as shown in Fig. 2(a) and (b). During training with sufficient samples, the problem of large-scale variations of RS objects can be compensated by using multiscale training [28] strategies and FPN [29] components. However, the scale distribution of RS objects is rather sparse under the condition of few-shot learning. A few annotated samples make it difficult to reflect the intraclass scale distribution characteristics. For example, we display the scale distribution maps of the dam and train station categories in Fig. 3, where green represents all samples and orange represents the few-shot samples. It is obvious to see that the few-shot data are too scarce to reflect the real scale distribution. Therefore, the locating ability of objects will be affected due to the insufficient scale information.

B. Small Interclass Feature Differences

Considering that RS images are obtained through overhead shooting, the structural characteristics of the object can be captured. If different categories behave similarly in terms of shape, color, and texture, misjudgments are likely to occur. Especially, when only a few samples are participating in training, the confusion among different RS classes will be intensified. In Fig. 2(c) and (d), we compare overpasses with bridges, and expressway toll stations with train stations, which are similar in structure and appearance. Using only a few training samples may cause the classes to be confusingly

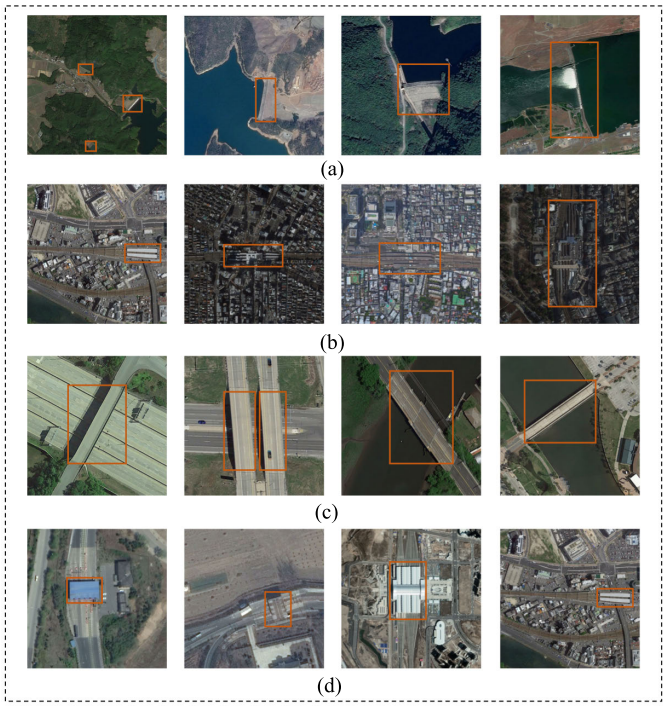


Fig. 2. Analysis of RS objects. (a) Dam objects at different scales. (b) Train station objects at different scales. (c) Comparison of overpasses versus bridges. (d) Comparison of expressway toll stations versus train stations.

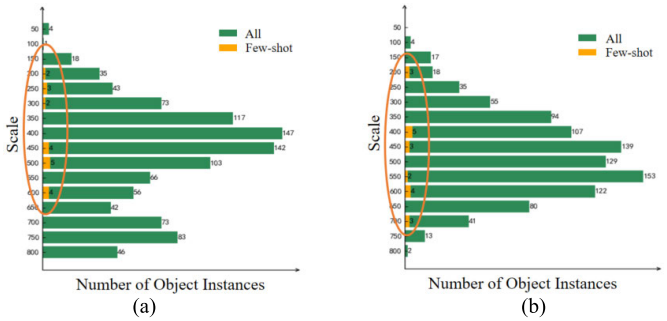


Fig. 3. Scale distribution of all objects and few-shot objects of the dam and train station categories on DIOR. We use green to represent the number of all objects and orange to represent the number of few-shot objects. (a) Scale distribution of dam. (b) Scale distribution of train station.

distributed. Therefore, it is essential to boost the classification capabilities of both base and novel classes.

Considering the issues outlined above, we introduce an FSOD method tailored for RS imagery, referred to as the RoI feature refinement (RIFR) method. It can simultaneously optimize base and novel classes through a simple fine-tuning strategy. The method can improve the feature representation capabilities of different categories from the source without extending the network structure in parallel. Meanwhile, we design two key components: a scale-aware feature compensation module (SAFCM) and a prototype trihard (PT) loss. They can work collaboratively to effectively locate and classify objects belonging to base and novel classes.

Concretely, we propose a more powerful training pipeline that maintains the model's feature extraction capabilities by freezing the feature extractor and FPN parameters during the few-shot training stage. At the same time, we can increase the ratio of positive and negative samples by reducing the number of proposed objects. In this case, the background objects can

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mostly be filtered out and foreground objects can receive more attention during network training. To address the sparse distribution of object scales within a class, we propose the SAFCM. In the two-stage detection network, RoI features are acquired by performing RoI pooling [30]/RoI align [31] on a specified feature map according to the hierarchical assignment rule. In order to make up for the limited representation ability of single-scale RoI features, we obtain RoI features corresponding to different scale feature maps. By compensating for insufficient scale information, the network's ability to perceive scale variations of the same class objects is enhanced, thereby greatly boosting the localization ability of novel classes. Furthermore, the severe confusion among classes is another important factor restricting the performance of FSOD. To better distinguish different classes, we propose the PT loss. It aims to drive the RoI classification features of each sample to converge with its corresponding class prototype while staying away from other class prototypes. Compared with the common InfoNCE loss [32] in contrastive learning, the PT loss can effectively avoid the over-fitting problem by only constraining the most confusing categories instead of all categories. Meanwhile, the PT loss can control the optimization degree of the network by adjusting the margin value. Under this constraint, the FSOD network can better distinguish different classes, thereby effectively alleviating the occurrence of classification confusion between base and novel classes. Through the detailed experimental verification, our RIFR method can achieve advanced results on DIOR [33], NWPU VHR-10.v2 [34], and FAIR1M-Airplane [24] datasets.

In summary, this article contributes in the following ways.

1) We propose an FSOD method suitable for RS imagery, called RIFR. It can better balance attention to base and novel categories by using a unified and effective training pipeline, and make them reach a higher level concurrently. Our approach consistently achieves state-of-the-art results across three datasets.
2) Aiming at solving the issue of large intraclass scale variations, we design the SAFCM. By leveraging multiscale RoI features to compensate for insufficient single-scale RoI features, the localization ability of the novel classes can be promoted.
3) To address the problem of small interclass feature differences, we propose the PT loss. By strategically aligning the RoI classification features with their corresponding class prototypes while keeping them away from other class prototypes, the classification confusion phenomenon of base and novel classes is mitigated.

II. Related Works

A. Object Detection

Object detection is a primary task within the realm of computer vision. In general, the methods are classified as one-stage detector [35], [36], [37], [38] and two-stage detector [30], [39], [40], [41], [42], [43]. The former directly predicts the objects' categories and bounding boxes from a single network. Whereas the latter needs to first generate proposed boxes and then classify and localize them, typically having higher detection performance. Below we introduce some object detection branches related to this article, including general object detection, small object detection, and long-tail object detection.

1) General Object Detection: In recent years, numerous object detection approaches for RS images have been proposed, leading to substantial advancements. Dong et al. [1] construct the dataset WHU-RSONE for RS object detection and propose the CNN-SOSF method with suitable RoI scales. Zhang et al. [44] put forward the MSFC-Net, which includes a CSFF architecture and FC regression layer. It can better describe the complex semantics in RS scenes and capture objects with irregular aspect ratios. Combining the spatial distribution characteristics of RS objects, Yu and Ji [2] design the object-based BBI method called RSADet. Sun et al. [45] propose the PBNet method to capture contextual associations of complex composite objects in RS images while achieving object localization. Wei et al. [46] propose an oriented objects representation method called pairs of middle lines and present the O^2-DNet method to promote the detection of oriented objects. Tian et al. [5] design the EGAT to comprehensively capture multilevel relationships among objects, thereby boosting the network's effectiveness. Liu et al. [47] present the adaptive balanced network, which includes EECA, AFPN, and CEM modules. This method can alleviate the complicated background of RS images to a certain extent.

2) Small Object Detection: Compared with the natural scene images, RS objects have large-scale variations and are densely distributed. How to promote the location ability of small objects in RS images is particularly important. Wang et al. [48] build an ASFP module to obtain rich contextual information and propose an area-weighted loss to focus on small objects. Dong et al. [49] replace the original NMS with Sig-NMS to avoid filtering out small objects directly and use transfer learning to improve the annotation quality. GLSANet [50] enhances the semantic information from a global perspective and highlights the representation of small objects through a local attention pyramid. Rabbi et al. [51] put forward a sparse outlook network named SOLO-Net, in which the SOLO-PAN and Sigmoid-IoU loss can effectively alleviate the problem of small objects being overwhelmed. Biswas and Tešic [52] propose a small object difficulty modeling method, which can better locate small objects via heatmap-based region proposals. The custom focal loss they designed can pay more attention to difficult samples in images. On this method basis, they jointly use local and global contrastive losses to achieve the goal of domain adaptation [53].

3) Long-Tail Object Detection: In the real world, the class imbalance is a common problem, and long-tail object detection has emerged from it. Yang et al. [9] propose SLT-Net for ship long-tailed detection, featuring RestLET to maintain tail class-specific parameters and MI loss to promote distribution consistency. Zhang et al. [54] consider the interclass imbalance problem needs to be taken seriously and propose a novel sampling strategy. Li et al. [55] design a balanced group softmax module to balance the classifiers through groupwise training. A CDAF loss is proposed to dynamically control the contribution of different samples [56], where the calculation

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of the adjustment factor is related to the sample features. The AHRL method decomposes the feature space into a hierarchical structure and uses a coarse-to-fine approach to distinguish different cluster features [57]. Qi et al. [58] propose BACL, employing FCBL to mitigate head class competition and FHM to enhance tail class diversity. ROG [59] combines object-level classification with global-level ranking, improving detection while preventing the omission of low-ranking objects. Zhao et al. [60] introduce LogN, a logit regularization method for self-calibrating object detectors. RichSem [61] utilizes extra data to learn rich semantic information and coarse locations, thereby improving class feature representation in long-tail scenarios.

The above most methods can yield impressive results for practical RS problems. However, the performance is still limited when samples are extremely scarce, which requires combining the few-shot learning strategies. Next, we will introduce FSOD methods in natural scenes and RS scenes, respectively.

B. Few-Shot Object Detection

As the name suggests, FSOD entails achieving the detection with limited training samples. Typically, FSOD methods are primarily categorized into two mainstream baselines: metalearning and fine-tuning learning. Meta R-CNN [14] represents a characteristic metalearning approach, acquiring knowledge of novel classes through the recombination of RoI features with class-specific attention vectors. Fan et al. [15] exploit the matching relationship between the support and query sets to capture richer foreground objects. Xiao et al. [16] propose a novel feature aggregation strategy to conduct FSOD and viewpoint estimation. Aiming to address the mismatching problem between RoI features and class prototypes, Han et al. [17] present the Meta-RPN method to detect old and new objects separately. Wang et al. [18] propose a fine-tuning approach (TFA) and find that only fine-tuning a small subset of parameters is more advantageous for detecting new categories. Qiao et al. [19] consider that it is important to decouple multistage and multitask, and they design the GDL and PCB modules to boost final detection. Sun et al. [20] propose the contrastive proposal encoding loss to cope with class confusion effectively. Jiang et al. [21] consider enhancing detection features from three perspectives, including spatial, task, and regularization. Fan et al. [12] mitigate forgetting by extending the network structure to combine detectors for base and novel classes. CFA [26] computes the gradients for both base and novel classes separately and then decides the ultimate direction of the gradient update by considering their respective angles. Li et al. [27] adopt a dual-branch structure to predict base and novel classes separately and merge the final results.

Currently, most FSOD methods only consider the detection performance of novel classes. Although several existing methods have taken into account the importance of detecting both base and novel classes simultaneously. They usually design different strategies to handle the prediction of base and novel classes, respectively, or retain base class knowledge by expanding the double branches. In contrast, we do not extend the network structure in parallel in the few-shot fine-tuning stage. Our method can learn discriminative features of different classes and achieve unified detection of base and novel classes. In this case, the model can also have better scalability, laying a good foundation for few-shot incremental object detection.

C. FSOD in RS Imagery

In contrast to natural scene imagery, RS imagery exhibits greater diversity and complexity, resulting in increased difficulty in FSOD. The quantity of FSOD methods suitable for RS scenes has been on the rise recently. FSODM [22] is a metalearning method, which deals with the problem of large-scale changes in RS images by extracting multiple feature layers. Aiming at solving the problem of the inconsistent direction of RS objects, Zhang et al. [62] present an oriented feature augmentation method to promote FSOD. Cheng et al. [23] propose a PLN to generate the prototypes with class-aware and use them to guide the generation of region proposals. To emphasize the spatial correlation, Zhang et al. [63] propose an SAGS module to maintain the local association of image features. Meanwhile, they come up with a TFS module and perform bidirectional attention to excavate the knowledge in the images. Lu et al. [24] propose to use the text-modal knowledge for FSOD. This method can better solve the problem of interclass confusion by introducing text features. Zhang et al. [25] argue that the performance of base classes is also crucial in FSOD, which has been ignored in most previous works. They design a two-branch structure to keep the memory of the model and an additional loss to realize separation among different classes.

The characteristics of RS images with large intraclass scale variations and small interclass feature differences will aggravate the challenge to the optimization of FSOD. Extremely few training samples for novel classes make it harder to optimize the network moderately and obtain discriminative features of RS objects. At the same time, existing methods do not take this into account too much, and they are prone to overfitting the lesser feature information, limiting the feature space of novel classes. Therefore, our proposed PT loss only adds constraints to the most confusing classes instead of full classes as existing methods do, thereby improving the model's generalization performance and avoiding overfitting problems. Furthermore, the SAFCM can boost the localization capability through a multiscale adaptive RoI features fusion strategy. In this case, our RIFR method is more robust both in localization and classification and can demonstrate superiority in balancing the detection of base and novel classes.

III. Methodology

In this session, we begin by providing a brief problem definition of FSOD to clarify our task. Next, we present an overview of the RIFR method. Then, we detail two innovative components, the SAFCM and PT loss. They can solve the regression and classification challenges in FSOD, respectively.

A. Problem Definition

In defining the problem of FSOD, we align with the settings established in the previous literature [14], [23]. All classes of

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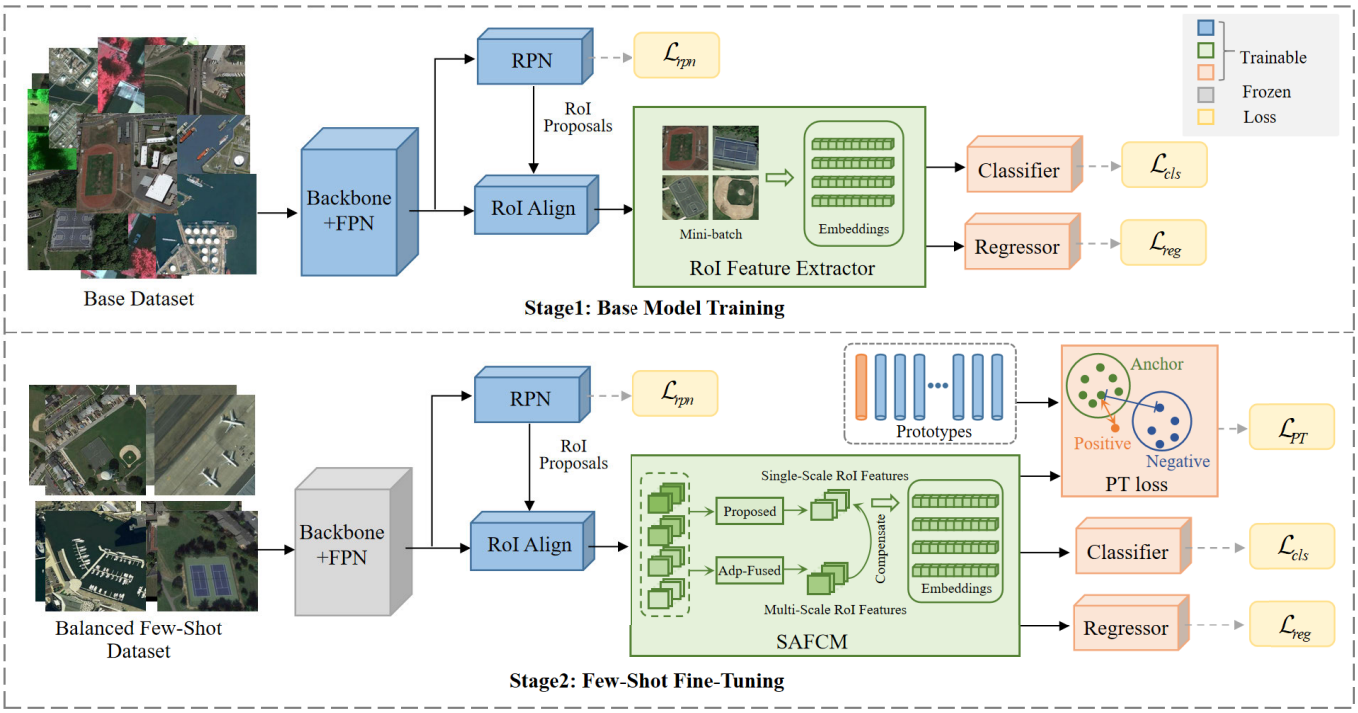


Fig. 4. Overarching structure of our RIFR method. The upper part represents the base model training network, and the lower part represents the few-shot fine-tuning network. We add SAFCM and PT loss, respectively, to facilitate the localization and classification of base and novel classes.

Algorithm 1 RIFR Method for FSOD in RS Imagery

Input: C_{base} with sufficient samples and C_{novel} with few-shot samples;

Output: Achieve object localization and classification on images containing both base and novel class objects;

Step 1: Base model Training
Train the base model using the base dataset;
Optimize the base model using $\mathcal{L}_{rpn}$, $\mathcal{L}_{cls}$, and $\mathcal{L}_{reg}$ loss functions;
Update the overall parameters of the base model;
Evaluate on test samples containing only base classes;
Step 2: Few-Shot Fine-Tuning
Integrate the SAFCM to the base model;
Fine-tune the base model using a balanced few-shot dataset;
Optimize the model using $\mathcal{L}_{rpn}$, $\mathcal{L}_{cls}$, $\mathcal{L}_{reg}$, and $\mathcal{L}_{PT}$ loss functions;
Update only part parameters of the model;
Evaluate on test samples containing all classes;

the entire dataset consist of base classes C_{base} and novel classes C_{novel}. They have no overlap $C_{\text{base}} \cap C_{\text{novel}} = \emptyset$. In detail, each category in C_{base} possesses adequate labeled samples, whereas each category in C_{novel} possesses only K instances annotations (e.g., $K = 3, 5, 10,$ and 20). Meanwhile, the complete set of labeled samples from C_{base} forms the base dataset. It can be denoted by $D_{\text{base}} = \{(x_b, y_b), y_b \in C_{\text{base}}\}$, where x_b denotes an image in the base dataset and y_b denotes its corresponding label. The few-shot dataset is composed of labeled samples of C_{novel}, which can be denoted by $D_{\text{novel}} = \{(x_n, y_n), y_n \in C_{\text{novel}}\}$. In addition to focusing on novel classes, we expect that base classes will not be sacrificed. Therefore, we will balance attention to base and novel classes, hoping that the model achieves better performance in all the learned classes.

B. Overview of Our RIFR Method

We propose the RIFR method for FSOD in RS imagery, as illustrated in Fig. 4. It is implemented utilizing the Faster R-CNN [30] with FPN [29] object detection framework. With the purpose of enhancing the network's object localization capability, we design the SAFCM. It utilizes multiscale RoI features to compensate for insufficient single-scale RoI features, thereby solving the problem of sparse distribution of few-shot data in scale. Furthermore, we design the PT loss to rectify the base and novel class RoI classification features, thereby suppressing the classification bias of the classifier. In recent years, metalearning methods have witnessed substantial advancements, leveraging information interaction between support and query samples. However, its shortcoming is that with the expansion of novel classes, the demand for storage space increases accordingly. In contrast, the training method based on the fine-tuning is simpler and more efficient. Therefore, our RIFR method adopts the fine-tuning training paradigm, which is specifically divided into the following two stages.

1) Base Model Training: We use base dataset D_{base} to update the entire network parameters. Since base classes contain sufficient labeled samples, the model can be fully optimized and achieve brilliant detection effects. During training, the loss remains consistent with the one defined in the Faster

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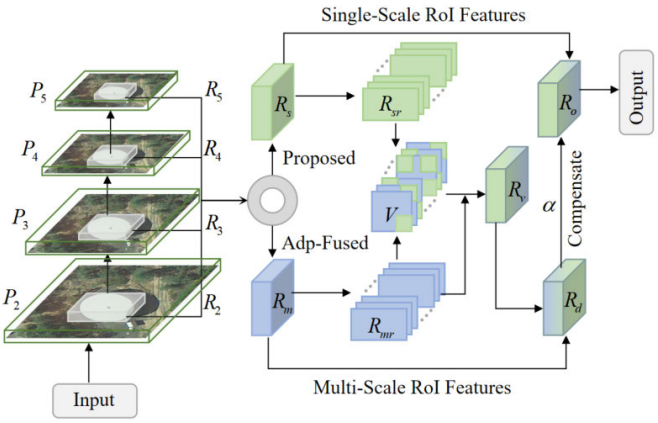


Fig. 5. Structure of the SAFCM. By using multiscale RoI features to make up for single-scale RoI features, the model can obtain more representative features and can better deal with the problem of scale sparse distribution in data.

R-CNN [30], which can be represented as follows:

$$\mathcal{L} = \mathcal{L}_{\text{rpn}} + \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} \tag{1}$$

where $\mathcal{L}_{\text{rpn}}$, $\mathcal{L}_{\text{cls}}$, and $\mathcal{L}_{\text{reg}}$ represent the foreground and background classification losses, the object's classification and regression loss, respectively.

2) Few-Shot Fine-Tuning: The training dataset comprises K instances for each category, referred to as the balanced few-shot dataset. We fine-tune the model with this dataset to prevent forgetting base classes when learning novel classes. TFA is a typical fine-tuning method in which only the last layer of the network undergoes parameter updates, while other parameters are kept frozen. Actually, it hinders the ability of the RPN and R-CNN components to perceive novel classes and makes the model underrepresent the novel classes. Therefore, we only freeze the backbone and FPN [29] to ensure the multiscale feature extraction capability of the model, while other components can be updated. Because of the weak ability of the network to perceive novel categories' objects, there are a large number of background objects in the proposed objects as negative samples. We adjust the ratio of positive and negative samples by reducing the original number of proposed objects, thereby increasing the focus on foreground objects. In addition, to boost the regression and classification capabilities of the network, we propose the SAFCM and the PT loss, with detailed explanations provided in Sections III-C and III-D, respectively. Furthermore, the loss function of our RIFR method in this stage can be expressed as follows:

$$\mathcal{L} = \mathcal{L}_{\text{rpn}} + \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} + \gamma \mathcal{L}_{\text{PT}} \tag{2}$$

where $\mathcal{L}_{\text{PT}}$ represents the PT loss that constrains the RoI classification features in network optimization. After adding the loss, the network can better distinguish different categories. The parameter γ represents the weighting factor, and we set the value of γ to 0.50 in the experiment. The processing of our method is summarized in Algorithm 1.

C. Scale-Aware Feature Compensation Module

As is widely acknowledged, FPN [29] is a potent module designed to address the challenge of substantial variations in object scales. It can increase the information association between contexts through top–down feature fusion operations. When there are enough training samples, features of different scales can be fully trained, and better performance can be achieved using FPN. However, it is not enough to only utilize FPN to deal with the large intraclass scale variances in FSOD of RS imagery. On the one hand, under the influence of imaging mode and observation distance, the scale range of RS objects is relatively larger. On the other hand, the number of novel classes' samples is extremely small. It is difficult for the model to capture the rich multiscale features with such few samples. In this case, the novel classes' objects may be missed.

Furthermore, when only FPN is used for multiscale feature learning, proposed RoI features are obtained by performing RoI Align on one scale feature map, and which feature map is specifically selected according to the RoI heuristical assignment rules. Small-sized objects correspond to lower level feature maps, while large-sized objects correspond to higher level feature maps. The RoI features generated in this way are only associated with a single-scale feature map, and their representation capabilities are limited. Especially when there are very few training samples, the ability of the network to perceive objects of novel classes is weak, and the final detection effect is unsatisfactory. Therefore, we propose the SAFCM based on FPN to compensate for the shortcomings of single-scale RoI features and thus obtain more discriminative features. It can fully utilize features at different scales and enhance the network's capacity to perceive variations in object scales. The structure of SAFCM is presented in Fig. 5. Specifically, we first obtain the RoI features corresponding to each scale based on the RPN [30]. Next, the multiscale RoI features are adaptively fused (adp-fused) to generate fused RoI features. Then, it is necessary to calculate the difference between the fused and the single-scale RoI features. Ultimately, the output is acquired by weighting the difference RoI features with the single-scale RoI features. The formulaic description of the module is as follows.

After an input image passes through the feature extractor and FPN, we can get a set of multiscale feature pyramids $\{\mathrm{P}_2, \mathrm{P}_3, \mathrm{P}_4, \mathrm{P}_5\}$. Then, the region proposal network can generate numerous proposal boxes under the action of anchor generators. According to the RoI heuristical assignment rules, each proposal box can only be assigned to a feature map and obtain the single-scale RoI features, which are denoted by $\mathrm{R}_s \in \{\mathrm{R}_2, \mathrm{R}_3, \mathrm{R}_4, \mathrm{R}_5\}$. However, in our method, we need to calculate the adaptive fused RoI features R_m for each proposal box corresponding to different scale feature maps

$$\mathrm{R}_m = \mathrm{Conv}(\mathrm{Cat}(\mathrm{R}_2, \mathrm{R}_3, \mathrm{R}_4, \mathrm{R}_5)) \tag{3}$$

where R_2, R_3, R_4, R_5, and $\mathrm{R}_m \in \mathbb{R}^{n\times c\times h\times w}$. n represents the count of proposal objects after filtering and c represents the channel count of the RoI features. Subsequently, we need to compute the disparity between the fused RoI features R_m and the single-scale RoI features R_s, which are named as the difference RoI features R_d. Its calculation process is as follows.

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According to R_s and R_m, we can calculate the related features R_v between them. First, we need to reshape R_s and R_m to $R_{sr} \in \mathbb{R}^{n\times c\times hw}$ and $R_{mr} \in \mathbb{R}^{n\times c\times hw}$. Then by multiplying R_{sr} and the transpose of R_{mr}, the correlation matrix $V \in \mathbb{R}^{n\times c\times c}$ between them can be obtained. Each element in it is expressed as v_{ij}:

$$v_{ij} = \frac{\exp\left(R_{sr}^i \cdot R_{mr}^j\right)}{\sum_{j=1}^{c} \exp\left(R_{sr}^i \cdot R_{mr}^j\right)}. \tag{4}$$

Also, the related features R_v between R_s and R_m is obtained by weighting R_m with the correlation coefficient v_{ij}

$$R_v^i = \sum_{j=1}^{c} v_{ij} R_m^j \tag{5}$$

where R_v^i denotes the related features of the ith channel and $1 \leqslant i \leqslant c$. Actually, the related features are realized by channel weighting of R_m, and the weighting coefficients are acquired by using R_s. In this case, we can gain similar parts between the single-scale and fused RoI features. Then, the difference RoI features R_d can be calculated through the difference operation of R_m and R_v

$$R_d = R_m - R_v. \tag{6}$$

The final output RoI features is obtained by weighting R_s and R_d, which can be expressed as follows:

$$R_o = R_s + \alpha R_d \tag{7}$$

where α represents the weighting coefficient. In the experiment, setting this coefficient to 0.10 can achieve better detection results.

D. PT Loss

Triplet loss [64] is first proposed for training face recognition models and achieves good results. A triplet consists of three samples, and each sample is represented by an embedding. They are referred to as the anchor, positive, and negative. Here, the anchor and positive samples pertain to the same class, whereas the anchor and negative samples correspond to different classes. The goal of this loss function is to encourage samples of the same category to cluster together, while concurrently urging samples from distinct categories to maintain a certain distance. Furthermore, to prevent different samples from aggregating into a very small space, the negative is farther away from the anchor than the positive by a margin. Therefore, the definition of the triplet loss can be expressed as follows:

$$\mathcal{L}_{\text{triple}} = \max[d(f(s_a), f(s_p)) - d(f(s_a), f(s_n)) + \text{margin}, 0] \tag{8}$$

where $f(s_a)$, $f(s_p)$, and $f(s_n)$ denote the embedding representations of the anchor, positive, and negative samples, respectively. The margin represents the boundary threshold for distinguishing positive and negative. Usually, the measuring distance function $d(\cdot)$ chooses the cosine similarity function.

When $d(f(s_a), f(s_n)) > d(f(s_a), f(s_p)) + \text{margin}$, the negative to the anchor is margin farther than the positive to the anchor. At this time, there is no need to optimize the loss function, and the triplets are called easy triplets. When $d(f(s_a), f(s_n)) < d(f(s_a), f(s_p))$, the negative to the anchor is closer than the positive to the anchor. Meanwhile, the triplets need to be paid more attention, they are called hard triplets. When $d(f(s_a), f(s_p)) < d(f(s_a), f(s_n)) < d(f(s_a), f(s_p)) + \text{margin}$, the positive and anchor is closer than the negative and the anchor, but the constraint of margin is not satisfied. They are called semihard triplets, and the loss value is smaller than hard triplets.

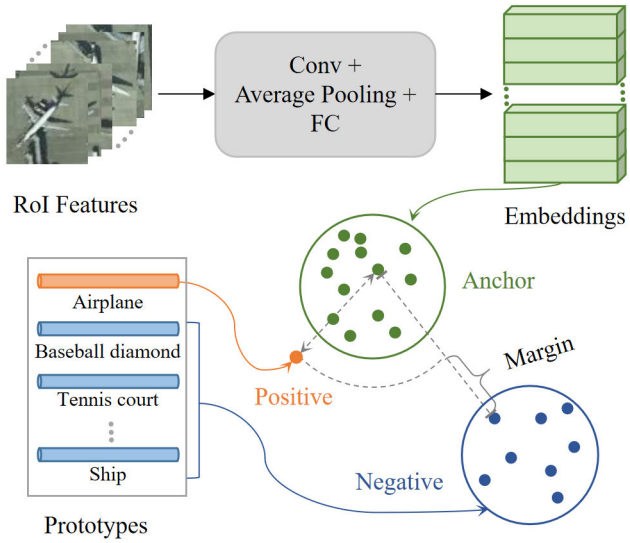


Fig. 6. Implementation of the PT loss. Under this constraint, the RoI classification features can converge with their class prototypes and stay away from other class prototypes.

From the above analysis, it can be seen that the effectiveness of the triplet loss depends on an appropriate triplet sampling strategy. Reasonable selection of sample combinations can help the model to better learn effective embedding representations. Therefore, when using the triple loss for metric learning, the sample selection and combination strategies also need to be carefully designed. In order to better distinguish different classes, we use a variant of triplet loss in our experiments called: trihard loss [65]. The advantage of trihard loss is that only the most easily confused positive and negative need to be selected. In few-shot learning, the loss can effectively promote the model to better distinguish between similar samples and dissimilar samples, avoid excessive optimization of the model, and improve the generalization performance of the model.

Through training, similar samples can be clustered more closely, and dissimilar samples can be more separated. The trihard loss can be expressed as follows:

$$\mathcal{L}_{\text{trihard}} = \max[\max(d(f(s_a), f(s_p))) - \min(d(f(s_a), f(s_n))) + \text{margin}, 0] \tag{9}$$

where $\max(d(f(s_a), f(s_p)))$ represents the distance from the most difficult positive to be classified to the anchor, and $\min(d(f(s_a), f(s_n)))$ represents the distance from the most difficult negative to be classified to the anchor. The margin also indicates the boundary threshold for distinguishing positive and negative.

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In the task of FSOD, we adopt the cosine similarity function for the final classification. Specifically, each proposed region will generate the corresponding RoI classification features, and each class will also have a corresponding classification weight, called the class prototype. In Fig. 6, we select the RoI classification features as the anchor, and select the positive and negative from the class prototypes to calculate the trihard loss. Therefore, we denote it as the PT loss. In detail, the output RoI features of the RPN network can be represented as $R_o \in \mathbb{R}^{n\times c\times w\times h}$. Then, it is necessary to execute the pooling operation on R_o to acquire the RoI feature vectors $E_o \in \mathbb{R}^{n\times c\times 1\times 1}$ for the classification and regression of objects. Similarly, n represents the count of proposed regions generated by the RPN after filtering, and c represents the channel count. In the calculation of PT loss, the RoI feature vector of each proposed region $E_o^k \in \mathbb{R}^{1\times c\times 1\times 1}$ is used as the anchor, that is, $f(s_a) = E_o^k$, and $1 \leqslant k \leqslant n$. The positive is determined as the class prototype to which the proposed object belongs. While the class prototype to which the proposed object does not belong is taken as the negative. This can be expressed as: $f(s_p) = w_p$ and $f(s_n) = w_n$. w_p and w_n are the class prototypes, and they are regarded as positive and negative, respectively. Therefore, the PT loss can be expressed as follows:

$$\mathcal{L}_{\mathrm{PT}} = \frac{1}{n}\sum_{k=1}^{n}\left\{\max\left[\max\left(d\left(E_o^k, w_p\right)\right) - \min\left(d\left(E_o^k, w_n\right)\right) + \text{margin}, 0\right]\right\}. \tag{10}$$

According to the PT loss rule, we perform loss calculation on the most confusing category of each proposed region, which effectively avoids the problem of novel categories being compressed into a smaller space, resulting in limited feature representation. Meanwhile, the samples of different classes can be separated from each other through iterative optimization, and the overfitting phenomenon can be effectively suppressed. In addition, the degree of constraint on the model can be adjusted by changing the margin value. In the experiments, we set the margin to 0.10 to achieve better results.

IV. Experiments

In this session, we not only introduce the experimental details but also conduct a comprehensive analysis of the experimental results. Through the quantitative and qualitative analyses, it is evident that our RIFR method attains excellent performance for FSOD in RS imagery.

A. Datasets

1) DIOR [33]: DIOR is a public dataset extensively used in the RS field. It contains 23 463 images collected from Google Earth, and the training, validation, and testing sets have 5862, 5863, and 11 738 images, respectively. All images in this dataset have a uniform size of 800 × 800, while the spatial resolution range is from 0.5 to 30 m. The entire dataset contains 20 classes. Following the previous settings in FSOD of RS images, we set 15 base classes and five novel classes. As shown in Table I, we perform four different splits on DIOR.

2) NWPU VHR-10.v2 [34]: NWPU VHR-10.v2 contains 1172 images collected from Google Earth and Vaihingen datasets. The training, validation, and testing sets are 679, 200, and 293 images, respectively. The uniform size of images in this dataset is 400 × 400, and the dataset contains a total of ten classes. There are seven classes that are set as base classes, and three classes are set as novel classes in FSOD.

3) FAIR1M-Airplane [24]: FAIR1M-Airplane is a fine-grained object detection dataset, which is proposed based on FAIR1M [66]. The dataset contains 18 377 images, and the training and testing sets have 12 964 and 5413 images, respectively. Same as [24], we do not set the corresponding validation set for it. All images are standardized to a size of 800 × 800. The dataset comprises 11 distinct airplane categories. Among them, the novel classes include A350, Boeing737, and Boeing747, while the rest are considered as base classes.

B. Evaluation Metrics

Generally speaking, the mean average precision (mAP) serves as the standard evaluation metric. It is the mean of average precision (AP) across all classes [67]. When evaluating, we need to set an IoU threshold to distinguish positive and negative samples, and it is selected as 0.5. According to the IoU threshold, we can calculate the true positives (TPs), false positives (FPs), and false negatives (FNs). Then, the precision (P) and recall (R) can be defined by them

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{11}$$

$$\mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}. \tag{12}$$

The relationship between precision and recall can be shown with a P–R diagram, and the area under it is defined as AP. Usually, AP is computed using the following approximation:

$$\mathrm{AP} = \sum_{\mathrm{R}\in(0,0.1,\ldots,1.0)} \mathrm{P}(\mathrm{R}). \tag{13}$$

The final mAP can be obtained by calculating the mean of each class AP. In our FSOD experiments, for a more comprehensive comparison, we not only evaluate the mAP of novel classes but also compare the mAP of base and all classes.

C. Implementation Details

The RIFR method we proposed is built upon Faster R-CNN [30], incorporating FPN [29]. Other comparison methods also use the same two-stage detection network to ensure the detection effect. This may be because the RPN network in Faster R-CNN can maximize the recall of more foreground objects and thus enable the objects of novel classes to be better captured. To make a fair comparison, we opt for the pretrained ResNet101 [68] as the backbone for various methods. There are two stages in the process of FSOD.

During the initial stage, we leverage the base dataset to train, thereby enhancing the model's feature extraction capabilities. We adopt the SGD optimizer and initialize the lr to 0.01. Also,

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TABLE I
SPLIT SETTINGS OF BASE AND NOVEL CLASSES ON DIFFERENT DATASETS

Datasets	Split	Novel					Base
DIOR	1	Baseball field	Basketball court	Bridge	Chimney	Ship	Rest
	2	Airplane	Airport	Expressway toll station	Harbor	Ground track field	Rest
	3	Dam	Golf course	Storage tank	Tennis court	Vehicle	Rest
	4	Express service area	Overpass	Stadium	Train station	Windmill	Rest
NWPU VHR-10.v2	–	Airplane	Baseball diamond	Tennis court	–	–	Rest
FAIR1M-Airplane	–	A350	Boeing737	Boeing747	–	–	Rest

TABLE II
FSOD RESULTS OF COMPARISON METHODS ON DIOR

Shot	Methods	Split1			Split2			Split3			Split4		
		Novel	Base	All	Novel	Base	All	Novel	Base	All	Novel	Base	All
3-shot	Meta-RCNN [14]	12.02	60.62	48.47	8.84	62.55	49.12	9.10	61.93	48.72	13.94	61.73	49.78
	FsDetView [16]	13.19	59.54	47.95	10.83	58.88	46.87	7.49	61.00	47.63	14.28	58.90	47.75
	P-CNN [23]	18.00	47.00	39.80	14.50	48.90	40.30	16.50	49.50	41.30	15.20	49.80	41.20
	TFA [18]	11.35	**70.32**	55.58	5.77	70.75	54.51	8.36	71.95	56.05	10.42	68.57	54.03
	DeRFCN [19]	28.25	61.90	53.48	14.55	61.41	49.70	15.78	63.56	51.47	10.83	59.81	47.50
	Retentive R-CNN [12]	21.39	69.05	57.14	11.18	69.61	55.00	13.62	70.46	56.25	12.67	69.44	55.25
	G-FSDet [25]	27.57	68.94	58.61	14.13	69.20	55.43	16.03	71.10	57.34	16.74	69.01	55.95
	RIFR (ours)	**28.51**	70.22	**59.79**	**15.11**	**70.83**	**56.90**	**17.78**	**72.16**	**58.57**	**18.22**	**69.10**	**56.38**
5-shot	Meta-RCNN [14]	13.09	62.01	49.78	10.88	63.14	50.07	12.29	63.44	50.66	15.84	62.60	50.91
	FsDetView [16]	14.29	58.58	47.51	9.63	60.31	47.64	12.61	61.33	49.15	15.95	58.97	48.22
	P-CNN [23]	22.80	48.40	42.00	14.90	49.10	40.60	18.80	49.90	42.10	17.50	49.90	41.80
	TFA [18]	11.57	70.51	55.78	8.19	70.79	55.14	10.13	71.64	56.26	14.29	68.85	55.21
	DeRFCN [19]	30.30	62.04	54.11	16.45	61.79	50.45	18.73	63.25	52.11	18.62	59.71	49.44
	Retentive R-CNN [12]	23.89	69.90	58.40	14.32	69.69	55.85	14.71	70.43	56.50	17.64	69.45	56.50
	G-FSDet [25]	30.52	69.52	59.72	15.84	69.25	55.87	23.25	70.18	58.43	21.03	67.96	56.30
	RIFR (ours)	**32.34**	**70.81**	**61.19**	**18.75**	**71.11**	**58.02**	**24.15**	**73.91**	**61.47**	**22.87**	**69.05**	**57.51**
10-shot	Meta-RCNN [14]	14.07	61.55	49.68	14.90	63.28	51.18	11.96	62.57	49.92	15.07	62.23	15.07
	FsDetView [16]	18.02	59.64	49.24	13.57	61.16	49.26	11.49	61.94	49.32	15.37	60.37	49.12
	P-CNN [23]	27.60	50.90	45.10	18.90	52.50	44.10	23.30	52.10	44.90	18.90	51.70	43.50
	TFA [18]	15.37	**70.52**	56.73	8.71	69.93	54.63	10.75	**72.56**	57.11	14.35	68.58	55.03
	DeRFCN [19]	32.64	61.51	54.29	18.40	61.42	50.69	20.43	64.55	53.52	21.61	59.54	49.99
	Retentive R-CNN [12]	27.73	69.03	58.71	14.90	69.49	55.84	19.11	70.56	57.70	17.78	69.35	56.46
	G-FSDet [25]	37.46	69.03	61.16	20.70	68.71	56.70	26.24	71.08	**59.87**	25.84	68.55	57.87
	RIFR (ours)	**37.74**	69.72	**61.73**	**21.93**	**70.77**	**58.56**	**26.46**	70.15	59.23	**28.49**	**69.22**	**59.03**
20-shot	Meta-RCNN [14]	14.45	63.21	51.02	16.71	63.86	52.07	16.14	65.53	53.18	18.17	63.24	51.98
	FsDetView [16]	18.01	62.69	51.52	14.76	61.16	49.56	17.02	65.17	53.14	16.96	60.89	49.91
	P-CNN [23]	29.60	52.20	46.80	22.80	51.60	44.40	28.80	53.10	47.00	25.70	52.30	45.70
	TFA [18]	17.96	**71.07**	57.79	12.18	70.02	55.56	17.99	**73.13**	59.35	12.01	**68.86**	54.65
	DeRFCN [19]	35.37	62.44	55.67	21.13	62.79	52.38	25.13	64.56	54.71	27.61	60.09	51.97
	Retentive R-CNN [12]	28.65	69.01	58.92	15.72	69.44	56.01	20.05	70.43	57.84	20.17	69.32	57.03
	G-FSDet [25]	39.83	69.80	62.31	22.69	68.18	56.86	**32.05**	71.26	61.46	31.78	67.73	58.75
	RIFR (ours)	**41.32**	70.96	**63.55**	**24.10**	**71.70**	**59.80**	29.76	72.13	**61.54**	**32.12**	68.78	**59.62**

The evaluation metric is mAP and the best results are painted in bold.

the model undergoes training for 36k iterations. In addition, the momentum is 0.9, and the weight decay is 0.0001. We can only learn the base classes at this stage. However, during the second stage, novel classes are required to be learned. We will use a balanced few-shot dataset to train the model, where each class only contains K-shot instances annotations and K is set to 3, 5, 10, and 20. Differently, the backbone and FPN are kept frozen to maintain the representation ability of the model. We also opt for the SGD optimizer and initialize the lr to 0.001. Same as [25], the model undergoes training for 6k iterations in the second stage. At the same time, due to the extreme scarcity of new categories' samples, the number of positive samples in the proposed objects regressed by the network is relatively small, while the background information are relatively large. Therefore, we reduce the number of proposals from the original 512–128 to achieve an appropriate ratio of positive and negative samples. In this case, the scarce foreground objects can receive more attention.

Meanwhile, we perform single-scale training on all three datasets. According to the image size, the input of the network is 800 × 800 for the DIOR and FAIR1M-Airplane datasets, while the input of the network is set to 400 × 400 for the rest dataset. The reported results are the average of five random experiments. Furthermore, all our experiments are conducted on two NVIDIA Tesla V100 GPUs.

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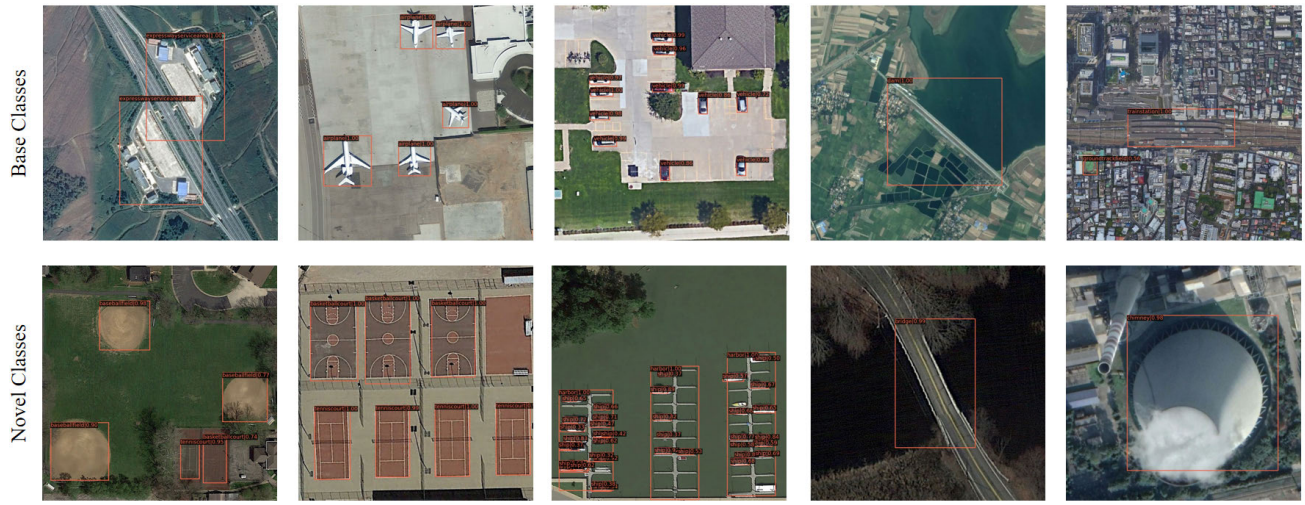


Fig. 7. Visualization of detection results in the ten-shot experiments for the first split on DIOR. The first row showcases base classes, while the second row showcases novel classes.

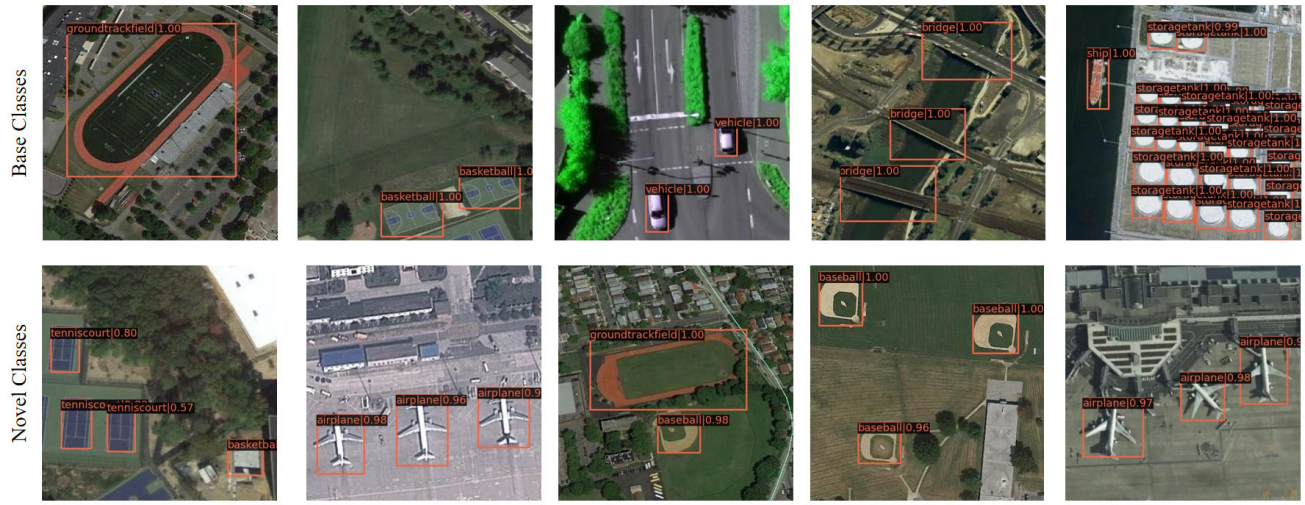


Fig. 8. Visualization of detection results in the ten-shot experiments on NWPU VHR-10.v2. The first row showcases base classes, while the second row showcases novel classes.

D. Experimental Results and Comparisons

To validate the efficacy of our RIFR method, we execute experiments on three datasets. Due to the split of base and novel classes exerting a pronounced influence on the experimental outcomes, we adopt the current mainstream split strategy for each dataset. Four kinds of splits are performed for the DIOR dataset, while only one kind of split is performed for the other two datasets. In the following, we will detail the performance of our RIFR method.

1) Results on DIOR: On DIOR, we conduct FSOD experiments of 3, 5, 10, and 20 shots. We record the mAP of novel, base, and all classes, respectively. Specifically, the evaluation metric for novel classes can mirror the transferability of the model to few-shot classes, while the evaluation metric for base classes can demonstrate the model's capability to retain information. The evaluation metric for all classes serves as an indicator for the overall learning performance during the FSOD phase. In addition, in terms of comparison methods, we not only compare Meta-RCNN, FsDetView, and P-CNN based on metalearning methods but also compare TFA, DeRFCN, Retentive R-CNN, and G-FSDet based on fine-tuning methods.

Concretely, the outcomes on DOIR are presented in Table II. The optimal results in each experiment have been marked in bold for more eye catching. From the experimental results, it is evident that while the metalearning methods attain certain effects on novel classes, they exhibit poor performance in base classes. In contrast, the methods based on fine-tuning can achieve higher scores for base classes. TFA is a typical FSOD method based on fine-tuning. The base class accuracy of this model is particularly high but the novel class accuracy may not be ideal. The DeRFCN method demonstrates superior results in natural scene imagery; however, its performance is not well when applied directly to RS imagery. Let us take the five-shot experiments of the first split as an example, it is 2.04% points lower than our method on novel classes and 7.08% points lower than our method on all classes. The retentive R-CNN and G-FSDet methods consider the performance of both novel

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TABLE III
FSOD RESULTS OF COMPARISON METHODS ON NWPU VHR-10.v2

Shot	Methods	Novel	Base	All
3-shot	Meta-RCNN [14]	20.51	87.00	67.05
	FsDetView [16]	24.56	87.68	68.75
	P-CNN [23]	41.80	82.84	70.53
	TFA [18]	8.80	89.35	65.19
	DeRFCN [19]	37.90	87.22	73.33
	Retentive R-CNN [12]	14.46	90.03	67.36
	G-FSDet [25]	49.05	89.11	77.01
	RIFR (ours)	**53.17**	**94.64**	**82.20**
5-shot	Meta-RCNN [14]	21.77	85.74	66.55
	FsDetView [16]	29.55	87.77	70.31
	P-CNN [23]	49.17	82.89	72.79
	TFA [18]	9.49	89.60	64.65
	DeRFCN [19]	46.08	86.87	74.63
	Retentive R-CNN [12]	20.29	90.01	69.09
	G-FSDet [25]	56.10	88.37	78.64
	RIFR (ours)	**62.39**	**95.26**	**85.40**
10-shot	Meta-RCNN [14]	26.98	87.01	69.00
	FsDetView [16]	31.77	87.75	70.96
	P-CNN [23]	63.29	83.05	78.11
	TFA [18]	9.26	89.95	65.74
	DeRFCN [19]	62.95	87.29	79.99
	Retentive R-CNN [12]	21.99	90.06	69.64
	G-FSDet [25]	71.82	88.40	83.43
	RIFR (ours)	**72.29**	**96.23**	**89.04**
20-shot	Meta-RCNN [14]	28.24	87.29	69.57
	FsDetView [16]	32.73	87.83	71.30
	P-CNN [23]	66.83	83.59	78.55
	TFA [18]	10.83	89.62	65.98
	DeRFCN [19]	64.61	87.26	80.46
	Retentive R-CNN [12]	26.57	90.62	71.41
	G-FSDet [25]	75.41	89.73	85.44
	RIFR (ours)	**79.46**	**95.38**	**90.61**

The evaluation metric is mAP and the best results are painted in bold.

TABLE IV
FSOD RESULTS OF COMPARISON METHODS ON FAIR1M-AIRPLANE

Shot	Methods	Novel	Base	All
3-shot	Meta-RCNN [14]	28.82	42.30	38.62
	FsDetView [16]	26.81	34.72	32.57
	TFA [18]	30.28	40.34	37.59
	FSCE [20]	26.16	39.71	36.02
	Retentive R-CNN [12]	28.82	40.64	37.42
	RIFR (ours)	**33.09**	**42.63**	**40.03**
5-shot	Meta-RCNN [14]	30.36	42.67	39.31
	FsDetView [16]	29.47	41.89	38.50
	TFA [18]	31.12	41.30	38.52
	FSCE [20]	31.80	40.66	38.24
	Retentive R-CNN [12]	31.50	40.53	38.06
	RIFR (ours)	**36.48**	**42.82**	**41.09**
10-shot	Meta-RCNN [14]	34.90	42.16	40.18
	FsDetView [16]	36.29	37.16	36.93
	TFA [18]	35.87	40.70	39.38
	FSCE [20]	34.94	41.72	39.87
	Retentive R-CNN [12]	31.96	40.66	38.28
	RIFR (ours)	**38.52**	**43.15**	**41.90**
20-shot	Meta-RCNN [14]	36.46	43.53	41.60
	FsDetView [16]	38.95	39.47	39.33
	TFA [18]	40.41	40.80	40.69
	FSCE [20]	39.36	41.15	40.66
	Retentive R-CNN [12]	32.04	40.59	38.26
	RIFR (ours)	**43.43**	**44.27**	**44.04**

The evaluation metric is mAP and the best results are painted in bold.

and base classes in their designs. Through the experimental results, we can see that the Retentive R-CNN method can maintain a good memory for base classes, but it has a poor learning ability for novel classes. G-FSDet is the existing top-performing FSOD method in the RS field, and it serves as our primary comparison method. Also, at the five-shot setting of the first split, we can see that our method exceeds G-FSDet by 1.82% points on novel classes and 1.47% points on all classes. More intuitively, we visualize the detection effects on DIOR in Fig. 7. The top row depicts the outcomes for base classes, while the bottom row portrays the outcomes for novel classes. Although there are several missed objects in the ship class. Overall, most classes perform well in terms of localization and classification.

2) Results on NWPU VHR-10.v2: We also carry out experiments on NWPU VHR-10.v2 to demonstrate the generalization performance of our method. The comparative results are presented in Table III. By comparing the results, it becomes apparent that utilizing metalearning methods does not lead to a significant reduction for base classes, while the novel classes do not exhibit outstanding results. In contrast, fine-tuning methods DeRFCN and G-FSDet can achieve decent scores in all classes. However, our RIFR method is still able to achieve outstanding advantages compared with other methods. For example, in the five-shot experiments, our method can outperform G-FSDet by 6.29% and 6.76% points on novel classes and all classes, respectively. When the number of few-shot objects reaches 20, our RIFR method can achieve 79.46% detection accuracy on novel classes and 95.38% on base classes. In addition, we visualize the detection effects on the dataset in Fig. 8.

3) Results on FAIR1M-Airplane: Moreover, we also assess the efficacy of the RIFR method on FAIR1M-Airplane. As far as the comparison methods are concerned, we choose the public methods Meta-RCNN, FsDetView, TFA, FSCE, and Retentive R-CNN. In Table IV, we show all the experimental outcomes. There is minimal disparity between metalearning and fine-tuning methods on this dataset, with our approach consistently achieving superior performance. In the five-shot experiments, our RIFR method can achieve 36.48% detection accuracy on novel classes, which is 4.68% points higher than the suboptimal FSCE method. When all classes are considered, the Meta-RCNN method outperforms FSCE by 1.07% points but is 1.78% points lower than our RIFR method. Simultaneously, as the count of objects increases to 20, the average accuracy of all classes can reach 44.04%. Similarly, we visualize the detection performance on the FAIR1M-Airplane dataset in Fig. 9.

E. Ablation Studies

Comprehensive ablation experiments are additionally conducted to showcase the effectiveness of our RIFR method. They are carried out under the ResNet101 backbone and the setting of the five-shot for the first split on DIOR. Since in FSL, the novel classes will receive more attention. Therefore,

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in addition to recording the overall mAP for novel, base, and all classes, we also record the mAP for each individual novel class. Because each category has a different sensitivity to parameter settings. Therefore, the overall performance of all classes is primarily taken into consideration when evaluating.

1) Ablation Study for Each Component: We present the contribution of each component in Table V. Our RIFR method adopts the fine-tuning training strategy, and a better baseline can be obtained by freezing the backbone and FPN during the second stage. Meanwhile, the baseline can achieve a 29.11% detection score on novel classes and 68.62% on base classes. To enhance the object localization and classification capabilities, we introduce SAFCM and PT loss to our method, respectively. By observing the experimental results of adding SAFCM, we can see that the accuracy improvement of the baseball field, basketball court, and chimney is more prominent. Compared with the baseline, they can improve the detection scores by 3.40%, 3.84%, and 2.43% points, respectively, while the class of bridge and ship can improve by 0.13% and 0.50% points, respectively. As we designed, SAFCM concentrates on the scale variations of objects. For novel classes, the baseball field, basketball court, and chimney have larger scale spans than the bridge and ship. Also, the PT loss is designed to deal with the confusion of base and novel classes. By adding additional loss constraints during the training process, the classification ability of all classes is boosted. Among them, the detection scores of the basketball court can be increased from the previous 35.86% to 40.70%. Not only the scores of novel classes has enhanced but the scores of base classes can also be increased from 68.62% to 70.60%. Therefore, the addition of PT loss can enable the model to obtain a better decision boundary. Ideally, our RIFR method can achieve superior performance when both SAFCM and PT loss are introduced. Both the novel and base classes can achieve the highest scores, and they are 3.23% and 2.19% points higher than the baseline, respectively. Moreover, the baseball field, basketball court, and chimney can also achieve the best performance, achieving scores of 45.68%, 42.44%, and 45.63%, respectively. To sum up, our RIFR method can maintain the memory for base classes while focusing on learning novel classes.

2) Ablation Study for the Backbone: As deep learning technology advances, foundation models are emerging as a popular trend, demonstrating notable success across various fields. We hope to explore what kind of gain effect it will bring to FSOD in RS imagery when the model is enlarged. Therefore, we use different backbones for our RIFR method, including ResNet101 [68], SwinV2-L [69], and SwinV2-L-RS, to conduct experiments respectively. In detail, the initial parameters of ResNet101 [68] and SwinV2-L [69] are publicly available, and they are pretrained on ImageNet [70]. However, SwinV2-L-RS are trained using two million RS data, and the training strategies are similar to RingMo [71]. Generally speaking, SwinV2-L pretrained on RS data performs better for all classes in Table VI. Although there is a certain decline in performance in certain ship and bridge categories, there is a breakthrough improvement in other novel categories. For example, the chimney category can be increased from 45.63% to 62.28%. The average performance of novel classes can be improved from 32.34% to 37.35%, which shows that RS foundation models are very beneficial to the FSOD task of RS images. Furthermore, the visualization of various models is depicted in Fig. 10. We observe some errors when utilizing ResNet101 as the backbone. For example, the localization for the basketball court category is not accurate, and there are many redundant object boxes around the objects. However, there are also errors in the classification of SwinV2-L-RS, it may classify the slender bridge as a dam. In summary, the more model parameters, the stronger the feature extraction ability. When performing few-shot learning, the foundation models can better represent various classes. At the same time, the models pretrained on RS data facilitate FSOD in RS imagery.

3) Ablation Study for the Parameter α in SAFCM: The parameter α is proposed in (7), which describes the weighting factor of difference RoI features. We also employ α to reflect the compensation degree from multiscale to single-scale features. The incorporation of multiscale information can improve the model's representational capability to a certain extent, particularly in scenarios, where data distribution is limited. However, when there is too much multiscale information, it may also cause certain interference to the detection.

It is observable that configuring α to 0.10 produces the best results for novel and all classes in Table VII. At this time, the score of novel classes is 31.17%, and the score of all classes is 60.02%. They are 2.06% and 1.28% points higher than the baseline, respectively. The highest accuracy on base classes can reach 69.66%, which is obtained with the α parameter set to 0.05. From this, we can see that when there is more multiscale feature guidance, it is not conducive to base class detection. Notably, different novel classes reach the optimal solution when the parameter α is set differently. The larger the α parameter is, the better it is for the classes with a large-scale variation. For example, the class of the ship can max out at the parameter of 0.05, while the baseball field can max out at the parameter of 0.20.

4) Ablation Study for the Margin in PT Loss: The margin represents the boundary distance that constrains positive and negative samples. As shown in (10), we need to set the size of the margin when calculating PT loss, and its value range is usually from 0 to 1. Through experimental verification, we observe that setting the margin to 0.10 yields the highest results in Table VIII. At this point, the accuracy of novel and base classes can reach 30.78% and 70.60%, respectively. In addition, when the margin is 0.10, the two classes of basketball court and bridge can also achieve better results. Also, they can be 4.44% and 1.48% points higher than the results when the margin is 0, respectively. However, the baseball field and ship categories can achieve better results when the margin is set to 0.05. They are 0.50% and 0.57% points higher than the results when the margin is 0, respectively. Noteworthily, the accuracy of novel classes is 30.08% when the margin is 0, which is higher than the baseline's accuracy of 29.11%. Also, when the value of the margin is greater than 0.10, the accuracy of all classes shows a downward trend. It follows that

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Fig. 9. Visualization of detection results in the ten-shot experiments on FAIR1M-Airplane. The first row showcases base classes, while the second row showcases novel classes.

TABLE V
ABLATION STUDY FOR DIFFERENT COMPONENTS ON DIOR

Methods	SAFCM	PT loss	Baseball field	Basketball court	Bridge	Chimney	Ship	Novel	Base	All
RIFR Method			41.78	35.86	15.16	39.74	13.03	29.11	68.62	58.74
	✓		45.18	39.70	15.29	42.17	**13.53**	31.17	69.64	60.02
		✓	43.45	40.70	**16.24**	40.53	12.99	30.78	70.60	60.64
	✓	✓	**45.68**	**42.44**	15.18	**45.63**	12.79	**32.34**	**70.81**	**61.19**

The evaluation metric is mAP and the best results are painted in bold.

TABLE VI
ABLATION STUDY FOR DIFFERENT BACKBONES ON DIOR

Methods	Backbones	Baseball field	Basketball court	Bridge	Chimney	Ship	Novel	Base	All
RIFR Method	ResNet101	45.68	42.44	15.18	45.63	**12.79**	32.34	70.81	61.19
	SwinV2-L	46.89	44.64	**16.33**	58.28	12.24	35.68	71.76	62.74
	SwinV2-L-RS	**52.76**	**48.52**	14.52	**62.28**	8.68	**37.35**	**71.78**	**63.18**

The evaluation metric is mAP and the best results are painted in bold.

TABLE VII
ABLATION STUDY FOR THE PARAMETER α IN SAFCM

α	Baseball field	Basketball court	Bridge	Chimney	Ship	Novel	Base	All
0.00	41.78	35.86	15.16	39.74	13.03	29.11	68.62	58.74
0.05	44.35	39.45	15.14	42.19	**14.06**	31.04	**69.66**	60.00
0.10	45.18	39.70	**15.29**	42.17	13.53	**31.17**	69.64	**60.02**
0.20	**45.79**	40.06	14.77	42.36	12.75	31.15	69.36	59.81
0.50	45.49	39.86	13.64	40.97	10.78	30.15	67.57	58.22
1.00	44.19	**42.70**	12.35	**43.38**	9.09	30.34	65.33	56.58

The evaluation metric is mAP and the best results are painted in bold.

when the margin constraint is too large, it is not conducive to distinguishing different categories.

5) Ablation Study for the Parameter γ in Loss Function: As shown in (2), the parameter γ represents the weighting factor of PT loss in the loss function. We consider that when the proportion of PT loss is too small, it may not have the effect of constraining the separability among classes. On the contrary, if the proportion of PT loss is too large, it may affect the normal progress of the detection task. It is evident that when γ is set to 0.50, all classes can achieve better performance in Table IX. The peak score on novel classes can reach 30.78%, whereas that on base classes can reach 70.60%. They are 1.67% and 1.98% points higher than the baseline, respectively. Furthermore, it is discernible from the table that the effect of various novel classes exhibits slight variations with changes in the parameter γ. It is well understood that the degree

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TABLE VIII
ABLATION STUDY FOR THE MARGIN IN PT LOSS

Margin	Baseball field	Basketball court	Bridge	Chimney	Ship	Novel	Base	All
0.00	42.96	36.26	14.76	**43.15**	13.27	30.08	68.43	58.84
0.05	**43.46**	39.00	15.46	41.50	**13.84**	30.65	69.13	59.51
0.10	43.45	**40.70**	**16.24**	40.53	12.99	**30.78**	**70.60**	**60.64**
0.30	42.82	38.76	15.99	37.08	12.17	29.36	69.42	59.40
0.50	41.21	36.53	15.67	35.40	11.30	28.02	68.26	58.20

The evaluation metric is mAP and the best results are painted in bold.

TABLE IX
ABLATION STUDY FOR THE PARAMETER γ IN LOSS FUNCTION

γ	Baseball field	Basketball court	Bridge	Chimney	Ship	Novel	Base	All
0.00	41.78	35.86	15.16	39.74	13.03	29.11	68.62	58.74
0.20	42.57	38.61	15.62	**40.98**	**14.07**	30.37	68.95	59.31
0.50	43.45	40.70	16.24	40.53	12.99	**30.78**	**70.60**	**60.64**
1.00	**43.62**	40.71	16.37	40.31	12.79	30.76	70.46	60.54
1.50	42.85	40.58	16.44	39.56	12.54	30.39	70.37	60.38
2.00	42.83	**40.86**	**16.68**	38.92	12.51	30.36	70.30	60.31

The evaluation metric is mAP and the best results are painted in bold.

TABLE X
ABLATION STUDY FOR DIFFERENT LOSS FUNCTIONS

Methods	Baseball field	Basketball court	Bridge	Chimney	Ship	Novel	Base	All
Baseline	41.78	35.86	15.16	39.74	**13.03**	29.11	68.62	58.74
+ InfoNCE loss	41.56	34.23	13.89	38.96	12.35	28.20	68.13	58.15
+ P-InfoNCE loss	42.23	37.02	14.96	39.11	11.96	29.06	69.02	59.03
+ Hinge loss	42.67	38.66	15.02	38.43	11.79	29.31	69.24	59.26
+ PT loss	**43.45**	**40.70**	**16.24**	**40.53**	12.99	**30.78**	**70.60**	**60.64**

The evaluation metric is mAP and the best results are painted in bold.

TABLE XI
ABLATION STUDY FOR THE NUMBER OF SHOTS

Shot	Baseball field	Basketball court	Bridge	Chimney	Ship	Novel	Base	All
3	35.82	41.05	9.86	46.44	9.41	28.51	70.22	59.79
5	45.68	42.44	15.18	45.63	12.79	32.34	70.81	61.19
10	45.50	51.43	17.80	59.33	14.63	37.74	69.72	61.73
20	47.30	51.38	17.66	65.26	25.02	41.32	70.96	63.55
50	56.26	59.91	18.07	67.90	21.30	44.69	70.15	63.78
100	61.60	65.23	24.66	68.69	26.05	49.24	71.08	65.62
200	63.27	68.55	25.56	71.17	27.86	51.28	71.93	66.77
500	**64.90**	**74.43**	**33.14**	**73.32**	**32.07**	**55.57**	**72.29**	**68.11**

The evaluation metric is mAP and the best results are painted in bold.

of confusion of different categories has certain differences, so they will reach their respective optimal solutions under different constraints. For example, when setting γ at 0.20 for chimney and ship classes, they can achieve detection scores of 40.98% and 14.07%, respectively. However, the class of baseball field, basketball court, and bridge may be more heavily confused with other classes, and thus achieve better results when the parameter γ is set larger. They can achieve detection scores of 43.62%, 40.86%, and 16.68%, respectively. Compared with the baseline, they can be increased by 1.84%, 5.00%, and 1.52% points, respectively.

6) Ablation Study for Different Loss Functions: To verify the effectiveness of PT loss, we compare it with other loss functions. InfoNCE loss [32] is commonly used in contrastive learning. Specifically, in a mini-batch, the proposed objects of the same category are considered positive samples, while objects of other categories are considered negative samples. As shown in Table X, the accuracy of novel classes is reduced from 29.11% to 28.20% after using InfoNCE loss. This suggests that there are certain limitations in directly using this loss in few-shot learning. On the one hand, only when the mini-batch size is set to a large value and the comparison categories are rich enough, a certain effect can be achieved. On the other hand, there are many background objects in the proposed objects, and comparing objects of a specific category with background information may not solve

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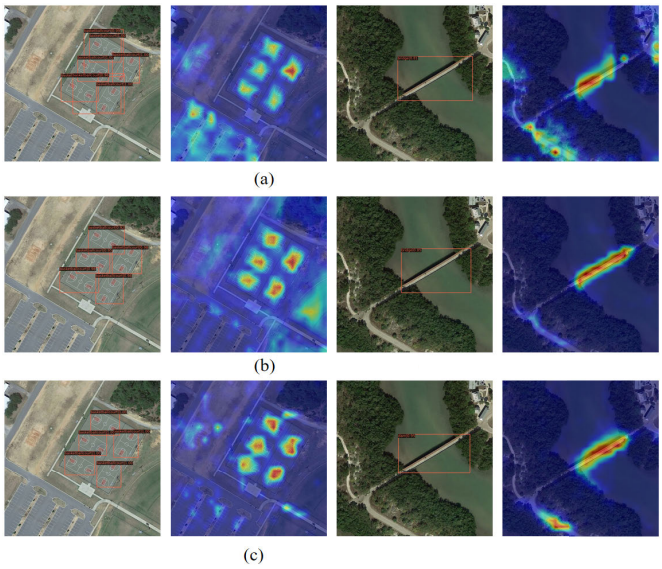


Fig. 10. Visual comparison of the RIFR with ResNet101, RIFR with SwinV2-L, and RIFR with SwinV2-L-RS. Their differences mainly lie in the use of different backbone networks and pretraining parameters. (a) RIFR (ResNet101). (b) RIFR (SwinV2-L). (c) RIFR (SwinV2-L-RS).

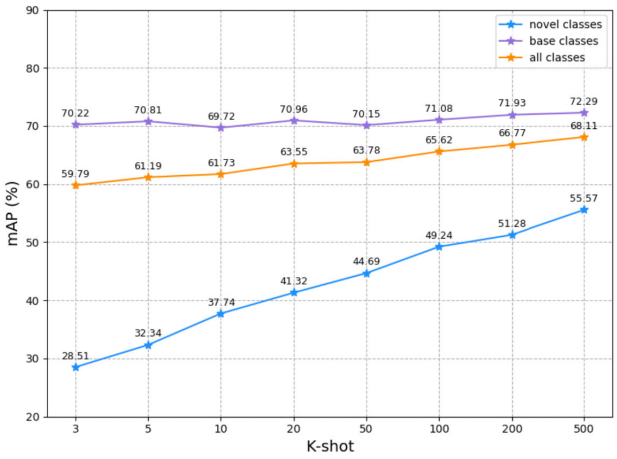


Fig. 11. mAP polylines of different classes with the number of shots increasing. Specifically, the blue line represents novel classes, the purple line represents base classes, and the orange line represents all classes.

the interclass confusion problem. To make the optimization of the objective function more straightforward, we modify InfoNCE loss and name it P-InfoNCE loss. The prototypes are the key to classification in few-shot learning. Therefore, we take the prototype of the same category for each proposed object is used as a positive sample, while the prototypes of other categories are used as negative samples. In this case, the accuracy of both the novel and base classes can be improved to a certain extent. In Hinge loss [72], we can adjust the degree of constraint on positive and negative samples by setting different margin values. Through ablation experiments, it can achieve a better experimental result than using P-InfoNCE loss.

On this basis, we hope to explore a more effective method. In our opinion, it is necessary to properly optimize the network to avoid overfitting in few-shot learning. In reality, a category is only heavily confused with one or a few specific categories. Therefore, we only need to constrain the most confusing category of each object as a negative sample instead of all other categories, which can improve the robustness of the model and avoid over-optimization problems. It can also be seen from Table X that compared with other loss functions, our proposed PT loss can achieve the best performance.

7) Ablation Study for the Number of Shots: We only list the experimental results when the shot is set to 3, 5, 10, and 20 in the comparative experiments. In this section, we wish to explore how higher performance can be achieved as the shot count increases. Therefore, we add more groups of experiments, where the shot is adjusted to 50, 100, 200, and 500 in Table XI. Not surprisingly, when the shot increases from 3 to 500, the accuracy of novel classes can be increased by 27.06% points. At the same time, the accuracy of all classes can also be improved from 59.79% under three-shot to 68.11% under 500 shots. More notably, the accuracy of base classes does not change much. It can reach 70.22% under three-shot and 72.29% under 500 shots. Visually, we also draw the mAP change polylines with the number of shots expansion in Fig. 11. The purple line represents the base classes, the blue line represents the novel classes, and the orange line represents all classes. It is obvious that with the number of shots increasing, the novel classes undergo a prominent improvement.

8) Qualitative Analysis: More intuitively, we qualitatively analyze the performance of SAFCM and PT loss. In Fig. 12, we present samples containing basketball courts with different scales. In the absence of SAFCM, the model has a weak perception of the basketball courts with a smaller scale, and there will be some omissions. When SAFCM is added, the rich scale information improves the multiscale perception ability of the model. So that more objects can be located. In addition, we also compared the distribution of RoI classification features with t-SNE [73] in the case of unused and used PT loss on NWPU VHR-10.v2. Specifically, one point represents one RoI classification feature, and one color represents one category in Fig. 13. After introducing the PT loss, different categories are more decentralized. Therefore, the classification capacity of the model can be effectively enhanced.

F. Further Analysis

1) Comparison of Long-Tail and FSOD: Long-tail object detection addresses class imbalance issues, where high-frequency classes typically outperform low-frequency ones. It usually employs balanced training strategies, such as data augmentation, sample balancing, and classifier calibration to enhance focus on low-frequency classes. FSOD is an extreme case of class imbalance, with a significant gap between base and novel classes, where each novel class has only a handful of training samples. Unlike the long-tail class distribution, where the amount of data in different classes decreases gradually, few-shot learning is oriented toward a more polarized class distribution. Achieving a satisfactory balance between base and novel classes is more challenging when directly applying long-tailed object detection methods to few-shot data. Since balanced training strategies in long-tailed object detection methods are more sensitive to sample size, novel classes risk overfitting due to limited samples, while

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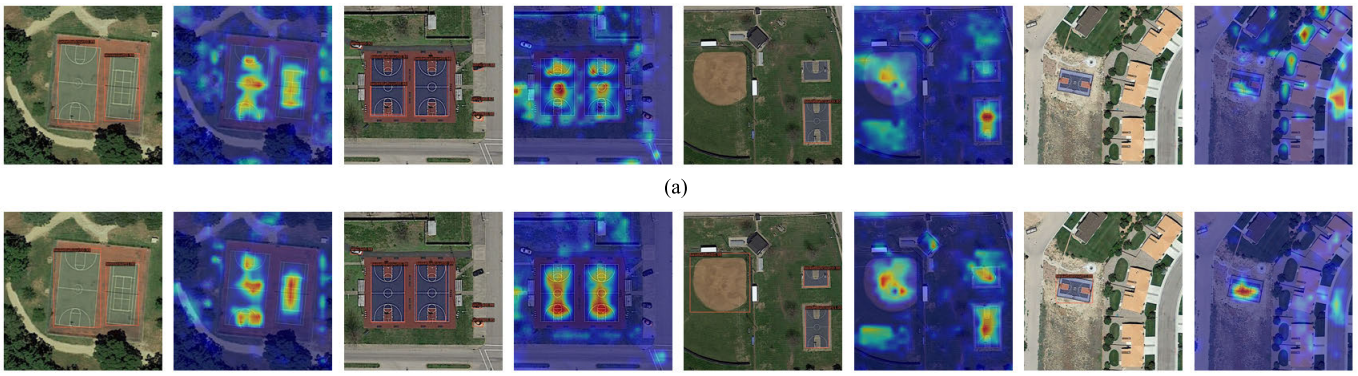


Fig. 12. Comparison of detection results and heat map visualization without and with SAFCM. The basketball courts on the left are larger, while the basketball courts on the right are smaller. When SAFCM is not added, some basketball courts with smaller scales are missed. (a) Detection results and visualization w/o SAFCM. (b) Detection results and visualization w/ SAFCM.

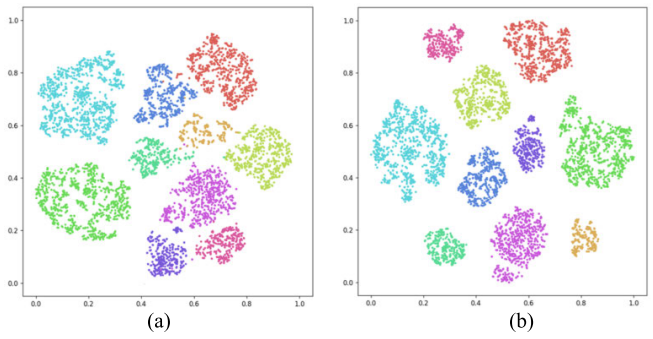


Fig. 13. Comparison of the distribution of RoI classification features without and with PT loss. We visualize them using t-SNE, where each dot represents an RoI feature and each color represents a category. (a) RoI features w/o PT loss. (b) RoI features w/ PT loss.

TABLE XII
COMPARISON OF LONG-TAIL AND FSOD METHODS

Methods	Novel	Base	All
BACL [58]	30.45	34.82	33.63
ROG [59]	16.53	30.53	26.71
LogN [60]	10.58	38.43	30.83
RichSem [61]	35.03	39.09	37.98
Meta-RCNN [14]	30.36	42.67	39.31
FsDetView [16]	29.47	41.89	38.50
TFA [18]	31.12	41.30	38.52
FSCE [20]	31.80	40.66	38.24
Retentive R-CNN [12]	31.50	40.53	38.06
RIFR (ours)	**36.48**	**42.82**	**41.09**

The evaluation metric is mAP and the best results are painted in bold.

base classes may suffer underfitting from insufficient focus. Thus, the extreme distribution in few-shot data necessitates a more refined balancing strategy, coupled with effective feature transfer methods. More intuitively, we evaluate recent long-tail object detection models, including BACL [58], ROG [59], LogN [60], and RichSem [61]. We present the experimental results on FAIR1M-Airplane under the five-shot setting in Table XII.

ROG considers the absence of low-ranking objects, but its classification capacity is limited, achieving only 26.71% accuracy on all classes. LogN effectively suppresses background objects through logit normalization but struggles with novel classes' classification, attaining only 10.58% accuracy for these classes. BACL increases novel class diversity and mitigates the numerical advantage of base classes, achieving 33.63% detection scores on all classes. RichSem learns from semantic information and coarse locations with extra textual and image data, reaching 37.98% detection scores on all classes. Despite the achievements of long-tailed object detection models, our proposed RIFR method significantly outperforms them, excelling in both novel and base classes. RIFR is not only more suitable for few-shot data with extreme distributions but also takes into account the intrinsic characteristics of RS objects. In conclusion, the primary distinction between long-tail and FSOD lies in the data distributions they address, leading to differences in model design. Practical applications should consider the specific data distribution when selecting an appropriate method.

2) Analysis of Different Methods' Effectiveness and Efficiency: To make a multidimensional comparison of different methods, we also evaluate the models' parameters, GPU memory requirements (GPU-M), flops, and FPS. The comparison results are shown in Table XIII. The experiments are performed on FAIR1M-Airplane under the five-shot setting. Notably, Meta-RCNN and FsDetView are metalearning methods that need to establish connections between support and query images. Their process is relatively complex, leading to lower computational efficiency, with FPS only reaching 3.27 and 3.04, respectively. However, their original backbone network does not contain FPN, which reduces some parameters in terms of model structure. The number of parameters for Meta-RCNN is only 50.51 M. The other four methods are based on fine-tuning methods, and they do not differ much in model parameters. TFA and FSCE are primarily designed to improve the performance of novel classes, and the direct optimization objective makes them more efficient. The FPS of TFA can be as high as 9.48. In contrast, the retentive R-CNN method is designed with a double-branch detector to consider the detection of base and novel classes. This reduces the computational efficiency, with an FPS of 4.33.

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TABLE XIII
COMPARISON OF DIFFERENT METHODS' EFFECTIVENESS AND EFFICIENCY

Methods	mAP ↑			params ↓	GPU-M ↓	flops ↓	FPS ↑
	Novel	Base	All				
Meta-RCNN [14]	30.36	42.67	39.31	**50.51**	3219	243.89	3.27
FsDetView [16]	29.47	41.89	38.50	56.80	3395	264.56	3.04
TFA [18]	30.28	40.34	37.59	60.39	**1955**	**179.17**	**9.48**
FSCE [20]	31.80	40.66	38.24	61.57	2001	180.35	6.08
Retentive R-CNN [12]	31.50	40.53	38.06	60.33	2485	210.37	4.33
RIFR (ours)	**36.48**	**42.82**	**41.09**	61.88	2219	207.21	5.32

Our proposed RIFR method aims to enhance feature representation of different classes from the source, utilizing a single-branch structure that treats base and novel classes uniformly. To enhance the model's multi-scale feature representation and classification capabilities in few-shot scenarios, we introduce SAFCM and PT loss, which result in a negligible increase in model parameters. In comparison, it is slightly more computationally efficient than Retentive R-CNN, achieving an FPS of 5.32. To sum up, our method demonstrates outstanding performance advantages on both base and novel classes with comparable model parameters and better computational efficiency. It can achieve a better balance between detection effectiveness and efficiency.

V. CONCLUSION

In this article, we put forward a novel FSOD method for RS images, employing a strong fine-tuning training strategy, named the RIFR method. It uses a unified network architecture to learn different classes, achieving better performance on both base and novel classes. To address the challenge posed by large intraclass scale variations of RS objects, we design the SAFCM to make up for the lack of single-scale feature representation, thereby overcoming the missing detection of novel classes. Furthermore, to tackle the issue of interclass confusion of RS objects, we introduce the PT loss to make different classes easy to distinguish. Thorough experiments conducted on three datasets demonstrate the remarkable performance of our RIFR method. In the future, we aim to delve deeper into the potential applications of RS foundation models, considering their integration into the realm of zero-shot learning to broaden our research scope.

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